THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 7: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bag has 60 balls, red or blue. If $P(\text{red}) = \frac{7}{10}$, the number of blue balls is: 1

(a) 20 **(b)** 19 **(c)** 17 **(d)** 18

2. A letter is chosen from the word EXAMINATION. The probability that it is neither A nor N is: 1

(a) $\frac{4}{11}$ **(b)** $\frac{7}{12}$ **(c)** $\frac{8}{11}$ **(d)** $\frac{7}{11}$

3. The probability that a seed germinates is $\frac{5}{8}$. Out of 4800 seeds the expected number that will NOT germinate is: 1

(a) 1802 **(b)** 1801 **(c)** 1799 **(d)** 1800

4. If an experiment has n equally likely outcomes, the probabilities of all outcomes add up to: 1

(a) 1 **(b)** 0 **(c)** $\frac{1}{n}$ **(d)** n

5. Two dice are rolled together. The probability that the product of the numbers is 9 is: 1

(a) $\frac{1}{18}$ **(b)** $\frac{1}{36}$ **(c)** $\frac{1}{37}$ **(d)** $\frac{35}{36}$

6. Two dice are rolled together. The probability that the sum is divisible by 4 is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{4}$

7. Three coins are tossed. The probability of getting more heads than tails is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{3}$

8. A bag has 9 red balls and some blue balls. If the probability of drawing a blue ball is $\frac{4}{7}$, the number of blue balls is: 1

(a) 11 **(b)** 12 **(c)** 14 **(d)** 13

9. **Assertion (A):** When two fair coins are tossed together, the probability of getting exactly one head is $\frac{1}{2}$. 1

Reason (R): The sample space is $\{HH, HT, TH, TT\}$ with 4 equally likely outcomes, of which HT and TH are favourable.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A coin tossed 100 times shows heads 54 times, so the experimental probability of a head is 0.54. 1

Reason (R): The theoretical probability of a head for a fair coin is $\frac{1}{2}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the probability that the larger of the two numbers on rolled dice is exactly 5. 2

12. Two dice are rolled together. Find the probability that the product of the numbers is 5. 2

13. A bag has 18 red balls and some blue balls. If $P(\text{blue}) = \frac{4}{7}$, find the number of blue balls. 2

14. Find the probability that the larger of the two numbers on rolled dice is exactly 6. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bag contains red, blue and green balls. $P(\text{red}) = \frac{3}{12}$ and $P(\text{blue}) = \frac{4}{12}$ and there are 60 balls in all. Find $P(\text{green})$ and the number of balls of each colour. 3

16. In 80 throws of a die, the number of times a prime number appeared was 41 and a number greater than 4 appeared 25 times (these include 11 throws showing 5). Find the experimental probability of 3
(i) a prime number,
(ii) a number greater than 4,
(iii) a prime number or a number greater than 4.

17. A letter is chosen at random from the word CHANDIGARH. Find the probability that it is 3
(i) not H,
(ii) a vowel or H.

OR

A bag contains red, blue and green balls. $P(\text{red}) = \frac{2}{9}$ and $P(\text{blue}) = \frac{5}{9}$ and there are 54 balls in all. Find $P(\text{green})$ and the number of balls of each colour. 3

18. In 250 throws of a die, the number of times a prime number appeared was 130 and a number greater than 4 appeared 84 times (these include 39 throws showing 5). Find the experimental probability of 3
(i) a prime number,
(ii) a number greater than 4,
(iii) a prime number or a number greater than 4.

OR

A letter is chosen at random from the word RAJASTHAN. Find the probability that it is 3
(i) not A,
(ii) a vowel or A.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Two fair coins are tossed together in a school game. 5
(i) Write the sample space.
(ii) Find the probability of getting two heads.
(iii) Find the probability of getting at least one head.
(iv) Verify that $P(\text{at least one head}) = 1 - P(\text{no head})$, and find how many times exactly one head is expected if the pair is tossed 80 times.

20. Two dice are rolled together. Find the probability that 5
(i) the sum is 10,
(ii) the sum is more than 10,
(iii) the sum is at most 10. Verify that
(ii) and
(iii) add to 1.

OR

A bag has 8 red, 8 blue and 14 green balls. Some green balls are removed so that $P(\text{green})$ becomes $\frac{1}{5}$. 5
How many green balls were removed?

END OF QUESTION PAPER — Total: 40 Marks